Solving AAS and ASA Triangles with the Law of Sines Answer Key

Precalculus

Quick Answers

1. 75 degrees
2. 18.03
3. 8.86
4. 14.02
5. (a) angle B = 40 degrees, —
6. 21.40
7. 20.71
8. 309.01 m
9. 239.20 m
10. (a) distance a = 92.53 m, —

What is verified

10 of 12 answers machine-checked against SymPy, across 10 problems.
— marks an answer only you can judge — problems 5, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus
HSG-SRT.D.10 – *problems 1, 5*
HSG-SRT.D.11 – *problems 2, 3, 4, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 12 tagged checks.

Common wrong answers

2. *If they got 7.99: paired the side with the wrong angle, dividing by the sine of seventy-five instead of the sine of forty*
3. *If they got 8.15: used the given side with the sine of the angle it lies between instead of first finding the third angle*

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1. $A = 42°$, $B = 63°$, $c = 10$. Find C.

Solution: the three angles total $180°$, so no law of sines is needed for this one: $C = 180° - 42° - 63° = 75°$. The given side $c = 10$ is not used here; it fixes the size of the triangle, not its angles. $\boxed{C = 75°}$

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2. $A = 40°$, $B = 75°$, $a = 12$. Find b.

Solution: a is paired with A and b with B, and both angles are known, so the law of sines applies directly — no third angle needed.

$$\frac{b}{\sin 75^\circ} = \frac{12}{\sin 40^\circ} \quad \Longrightarrow \quad b = \frac{12\sin 75^\circ}{\sin 40^\circ} = \frac{11.5911\ldots}{0.6428\ldots} = 18.0305\ldots$$

$\boxed{b \approx 18.03}$

Common wrong answer: 7.99, from writing the ratio upside down — $12\sin 40^\circ/\sin 75^\circ$. The side you want goes on top with *its own* angle underneath.

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3. $A = 35^\circ$, $B = 80^\circ$, $c = 14$. Find a.

Solution: the known side is c, but C is not given, so find it first: $C = 180^\circ - 35^\circ - 80^\circ = 65^\circ$. Now c has a partner.

$$a = \frac{14\sin 35^\circ}{\sin 65^\circ} = \frac{8.0301\ldots}{0.9063\ldots} = 8.8603\ldots$$

$\boxed{a \approx 8.86}$

Common wrong answer: 8.15, from pairing $c = 14$ with the 80° angle it happens to sit next to. A side pairs with the angle *opposite* it, never an adjacent one.

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4. $B = 28^\circ$, $C = 47^\circ$, $b = 9$. Find c.

Solution: b pairs with B and c pairs with C, and both are known.

$$c = \frac{9\sin 47^\circ}{\sin 28^\circ} = \frac{6.5822\ldots}{0.4695\ldots} = 14.0204\ldots$$

Since $C > B$, the answer must be longer than 9, which it is. $\boxed{c \approx 14.02}$

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5. $A = 22^\circ$, $C = 118^\circ$, $a = 7$. (a) Find B. (b) Why is the triangle determined?

(a) $B = 180^\circ - 22^\circ - 118^\circ = 40^\circ$. $\boxed{B = 40^\circ}$

(b) *Instructor-judged.* A correct response says the angle sum fixes the third angle, so all three are known and the *shape* is settled, and the one given side then fixes the *scale* through the law of sines — every other side follows from it. Two angles alone give a family of similar triangles; the side selects one member. Restating the angle sum without the scaling role of the side is partial credit.

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6. $B = 52^\circ$, $C = 61^\circ$, $a = 25$. Find b.

Solution: the known side is a, so A is needed: $A = 180^\circ - 52^\circ - 61^\circ = 67^\circ$.

$$b = \frac{25\sin 52^\circ}{\sin 67^\circ} = \frac{19.7003\ldots}{0.9205\ldots} = 21.4016\ldots$$

$\boxed{b \approx 21.40}$

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7. $A = 31°$, $C = 96°$, $c = 40$. Find a.

Solution: c is paired with the known C, and a with the known A, so go straight across.

$$a = \frac{40 \sin 31°}{\sin 96°} = \frac{20.6015\ldots}{0.9945\ldots} = 20.7150\ldots$$

$$\boxed{a \approx 20.71}$$

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8. Surveyors: $c = 500$ m apart, $A = 38°$, $B = 47°$. Find $a = BC$.

Solution: the baseline c is the side between the two measured angles, so this is ASA and the third angle comes first: $C = 180° - 38° - 47° = 95°$. The mast sits at C, and the distance from B to the mast is the side opposite A.

$$a = \frac{500 \sin 38°}{\sin 95°} = \frac{307.8181\ldots}{0.9962\ldots} = 309.0066\ldots$$

$$\boxed{a \approx 309.01 \text{ m}}$$

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9. Path: $b = 180$ m from bench A to fountain C, $A = 25°$, $C = 110°$. Find $c = AB$.

Solution: the known side b runs between the two measured angles, so find B first: $B = 180° - 25° - 110° = 45°$.

$$c = \frac{180 \sin 110°}{\sin 45°} = \frac{169.1447\ldots}{0.7071\ldots} = 239.2067\ldots$$

The answer is longer than the path itself, as it must be since C is the largest angle.

$$\boxed{c \approx 239.20 \text{ m}}$$

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10. Fire towers: $c = 150$ m apart, $A = 33°$, $B = 29°$. (a) Find $a = BC$. (b) Why is the triangle unique?

(a) $C = 180° - 33° - 29° = 118°$, and the distance from tower B to the smoke is the side opposite A.

$$a = \frac{150 \sin 33°}{\sin 118°} = \frac{81.6959\ldots}{0.8829\ldots} = 92.5263\ldots$$

$$\boxed{a \approx 92.53 \text{ m}}$$

(b) *Instructor-judged.* A correct response contrasts this with the two-sides-and-a-non-included-angle case: here the angle sum hands over the third angle at once, leaving no choice of shape, so exactly one triangle exists. With two sides and a non-included angle the unknown angle comes from an inverse sine, which offers an acute value *and* its obtuse partner, and both can produce a valid triangle. Full credit requires naming the inverse sine and its two candidate angles; ”this one is easier” is not enough.